

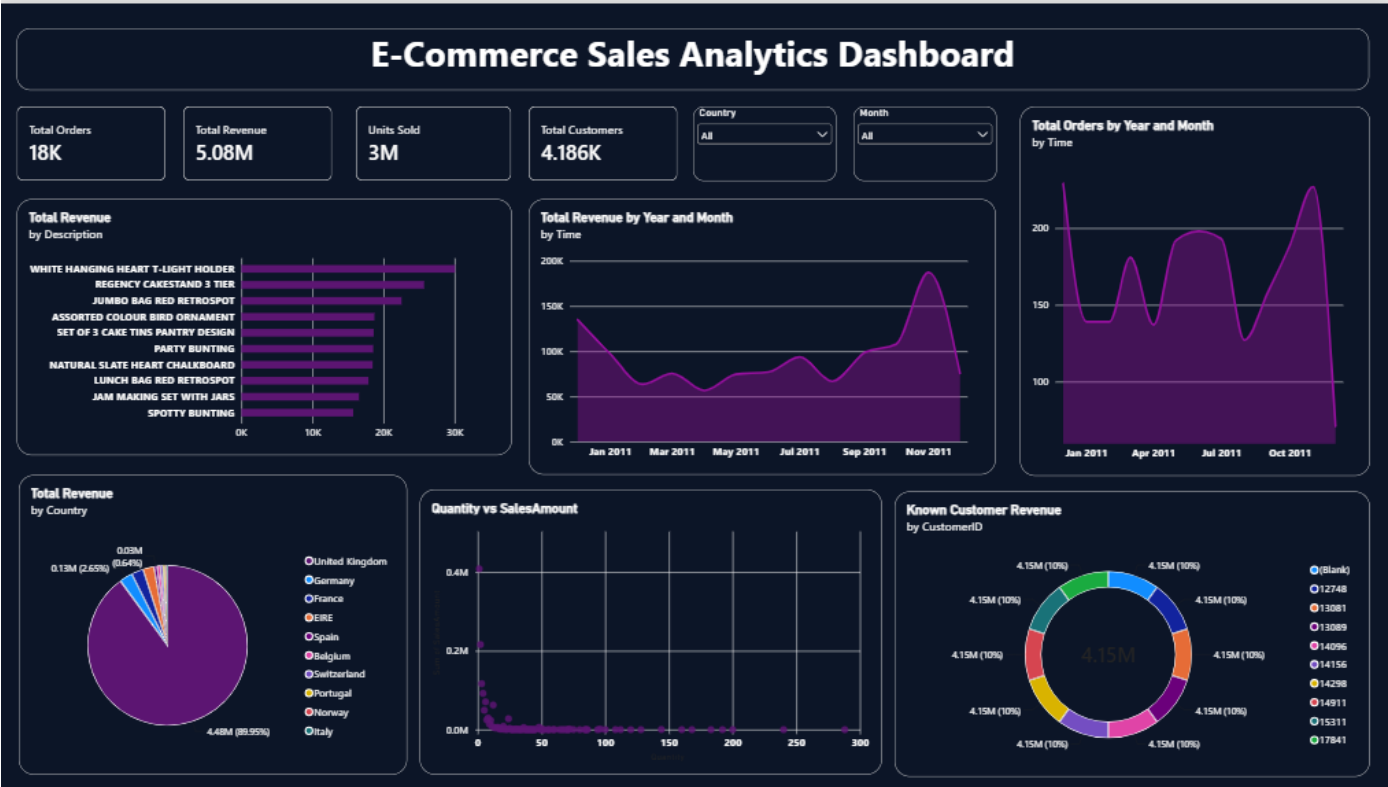
ONLINE RETAIL | EXECUTIVE ANALYTICS REPORT

ApexPlanet Data Analytics Internship • Debdut Nandy • September 2026

Executive summary. This analysis examines transactional e-commerce data from December 2010 to December 2011. Following cleaning and outlier filtering, 482,254 records were retained for sales analysis. The project combines exploratory analysis, SQL, Power BI, customer segmentation, revenue forecasting and an automated reporting pipeline.

£5.08M	18,088	3.02M	4,185	£281.05
TOTAL REVENUE	ORDERS	UNITS SOLD	KNOWN CUSTOMERS	AVG. ORDER VALUE

POWER BI DASHBOARD



Dashboard: revenue and order trends, product and country performance, quantity vs. sales, and customer revenue.

FIVE EVIDENCE-BASED BUSINESS INSIGHTS

- 1. UK concentration: The United Kingdom contributed £4.48M (88.2%) of recorded revenue.
- 2. Seasonal peak: November 2011 produced £742,122, the highest full-month revenue in the dataset; September and October also showed stronger sales.
- 3. Leading product: White Hanging Heart T-Light Holder generated £30,042, the largest product-level revenue total.
- 4. Customer value: 4,185 identified customers were recorded; the mean order value was £281.05.
- 5. International demand: Germany (£132,087) and France (£120,277) were the largest non-UK markets.

Scope note: December 2011 ends on 9 December and should not be compared directly with complete months. Customers with missing IDs remain in overall sales KPIs but are excluded from the unique-customer count.

ADVANCED ANALYTICS & ACTION PLAN

From descriptive performance to customer strategy, revenue planning and reproducible reporting.

CUSTOMER SEGMENTATION | K-MEANS (4 CLUSTERS)

SEGMENT	PROFILE	POSSIBLE ACTION
Cluster 0	Recent, lower-spending customers	Encourage a second purchase with relevant cross-sell offers.
Cluster 1	Recent, frequent, high-spending customers	Prioritize retention and loyalty recognition.
Cluster 2	Inactive, lower-spending customers	Test targeted reactivation and measure incremental response.
Cluster 3	Occasional customers	Use product-based reminders and repeat-purchase incentives.

DAILY REVENUE FORECASTING

A linear regression model used lagged revenue (previous day and previous week), a seven-day rolling average, day of week and month. Held-out evaluation produced:

£6,333.93	£7,868.82	0.5060
MAE	RMSE	R ²

Interpretation: the model explains approximately 50.6% of variance in the evaluation data. Forecasts are useful as directional planning inputs, not guaranteed sales outcomes.

THREE BUSINESS RECOMMENDATIONS

- 1. Plan inventory around observed seasonality. Prepare for elevated autumn demand, particularly ahead of November; confirm patterns using additional years before setting long-term targets.
- 2. Diversify market exposure. Maintain service quality in the UK while testing targeted growth initiatives in Germany and France, with country-level profitability tracked separately.
- 3. Tailor customer engagement by segment. Retain high-value repeat buyers, develop recent lower-spending customers, and evaluate reactivation campaigns against measurable uplift.

REPRODUCIBLE AUTOMATION & DELIVERABLES

GitHub Actions runs the Python cleaning and KPI pipeline on a weekly schedule and on manual demand. It installs dependencies, processes the source Excel file, verifies the cleaned CSV and Excel workbook, and uploads both as workflow artifacts. A successful manual run was verified in GitHub Actions.

Project repository: <https://github.com/Deb124-source/ApexPlanet-Task5-Debdut-Nandy>

Method note: Financial values are gross transaction-derived sales (Quantity × UnitPrice) after cleaning and IQR outlier filtering; they are not net profit. Source: UCI Online Retail transactional dataset; calculations: project automation workbook.